Function Spaces in Finite Element Methods

A Quick Intro to Hilbert, L^2, and Sobolev Spaces

Motivation: Why Special Function Spaces?

- In FEM, we often transform strong forms of PDEs (like $\nabla \cdot (k\nabla u) = f$) into **weak forms** using integration by parts.
- Weak forms look like: Find u such that
$$\int_\Omega (k\nabla u \cdot \nabla v d\Omega) = \int_\Omega f v d\Omega$$
for all suitable "test functions" v.
- This involves integrals of functions and their derivatives.
- **Question:** What kind of functions u and v can we "legally" plug into these integrals? Do they need to be smooth? Can they have kinks?

Analogy: From Vectors to Functions

- You know **vector spaces** like $\mathbb{R}^3$: vectors, addition, scalar multiplication.
- We measure vector "size" (norm) using the dot product:

$$||x||^2 = x_1^2 + x_2^2 + x_3^2 = x \cdot x$$

$$||x|| = \sqrt{x_1^2 + x_2^2 + x_3^2} = \sqrt{x \cdot x}$$

- Functions can also form vector spaces! We can add functions (`f+g`) and scale them (`c*f`).
- **Challenge:** How do we define a "dot product" and "size" (norm) for functions? How do we ensure our space is "complete" (no missing limits)?

Hilbert Spaces: The General Framework

- A **Hilbert Space** is a vector space (often of functions) with an **inner product** $\langle f, g \rangle$.
- The inner product lets us define:
- **Norm (Size):** $||f|| = \sqrt{\langle f, f \rangle}$
- **Orthogonality (Angle):** f is orthogonal to g if $\langle f, g \rangle = 0$.
- Crucially, Hilbert spaces are **complete**: Every Cauchy sequence converges to a point *within* the space . This is vital for proving solution existence and convergence of approximations (like FEM!).

$L^2(\Omega)$ Space: Square-Integrable Functions

- The most common Hilbert space in introductory FEM.
- **Definition:** $L^2(\Omega)$ is the space of functions `f` defined on a domain `Ω` whose square has a finite integral:

$$\int_\Omega |f(x)|^2 dx < \infty$$

- Think: Functions with finite "energy" or finite mean-square value.
- **Inner Product:**

$$< f, g >_{L^2} = \int_\Omega f(x) g(x) dx$$

- **Norm (L^2 Norm):**

$$||f||_{L^2} = \sqrt{\int_\Omega |f(x)|^2 dx}$$

- $L^2(\Omega)$ is a Hilbert space. It's also sometimes called $H^0(\Omega)$.

$L^2(\Omega)$ Examples

▶ Code

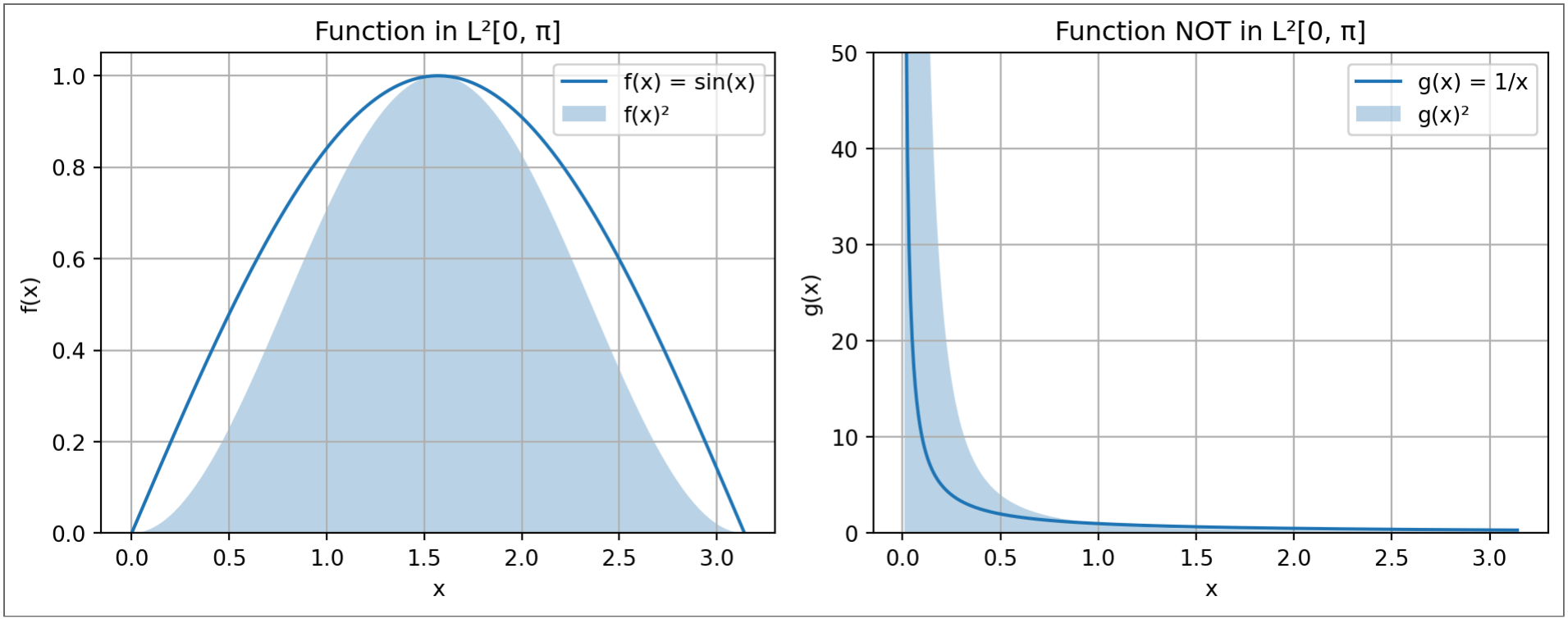

Figure 1: Left: sin(x) is in $L^2[0, \pi]$. Right: 1/x is NOT in $L^2[0, \pi]$ as its square diverges near x=0.

Note

- Functions can be unbounded ($1/\sqrt{x}$ near 0 on [0,1]) and still be in L^2.

- Functions can be discontinuous and be in L^2.

Need for Derivatives: Sobolev Spaces

- Our weak form:
$$\int_\Omega (k\nabla u \cdot \nabla v d\Omega) = \int_\Omega f v d\Omega$$
involves derivatives $\nabla u \cdot \nabla v$.
- Requiring functions u, v to be only in $L^2(\Omega)$ is not enough!
- We need their derivatives to also "behave well" (i.e., be square-integrable).

Problem

What if u is continuous but has a kink (like |x|)?

Its classical derivative isn't defined everywhere or might not be in L^2.

Sobolev Space $H^1(\Omega)$

- $H^1(\Omega)$ is the simplest Sobolev space relevant to many FEM problems.

Definition

$H^1(\Omega)$ is the space of functions f such that both f and its first weak derivatives are in $L^2(\Omega)$.
$H^1(\Omega) = \{f \in L^2(\Omega) \mid \nabla f \in L^2(\Omega)\}$

(Where ∇f denotes the vector of first weak partial derivatives).

- Weak Derivative: A generalization of the derivative using integration by parts. It allows functions with "corners" or "jumps" in their classical derivative (like $|x|$) to still have a derivative within this framework, provided it's an L^2 function.
- $H^1(\Omega)$ is also a Hilbert space.

$H^1(\Omega)$ Inner Product and Norm

- Inner Product: Includes terms for both the function and its derivative.
  $< f, g >_{H^1} = \int_\Omega f(x)g(x)dx + \int_\Omega \nabla f(x) \cdot \nabla g(x)dx$
  (Or sometimes weighted differently, but involves both parts).
- Norm (H^1 Norm): Measures both the size of the function and its derivative.
  $||f||^2_{H^1} = ||f||^2_{L^2} + ||\nabla f||^2_{L^2}$
- If $||f||_{H^1}$ is finite, the function is "well-behaved enough" for typical second-order PDE weak forms (like heat equation, elasticity).

$H^1(\Omega)$ vs $L^2(\Omega)$

▶ Code

Figure 2: Left: x^2 is smooth, both function and derivative (2x) are in $L^2[-1, 1]$, so $x^2 \in H^1[-1, 1]$. Right: Step function is in $L^2[-1, 1]$, but its derivative (a Dirac delta) is not an L^2 function, so Step $\notin H^1[-1, 1]$.

- Smooth functions (like polynomials) are typically in H^1.
- Functions with “jumps” (like the step function) are often in $L^2$$ but not H^1.

- Functions with “kinks” (like |x|) can be in H^1 because their weak derivative exists and is in L^2.
  $d/dx|x| = sign(x), \quad \text{which is in} \quad L^2[-1,1]$

Summary:

- Weak Formulations: FEM relies on integral (weak) forms of PDEs.
- $L^2(H^0)$: Guarantees that integrals of functions themselves ($\int f v dx$) make sense and are finite. Useful for source terms, mass matrices.
- H^1: Guarantees that integrals involving first derivatives ($\int \bar{\nabla} f \nabla v dx$) also make sense and are finite. This space is often the natural "solution space" for many second-order problems (heat, elasticity, potential flow). Ensures stiffness matrices are well-defined.
- Using these spaces provides a rigorous mathematical foundation for FEM, ensuring solutions exist and approximations converge.

Beyond H^1

- Higher-order Sobolev spaces exist, like $H^k(\Omega)$, which include functions with weak derivatives up to order k.
- Specific spaces, such as $H(\mathrm{div})$ and $H(\mathrm{curl})$, are useful for vector fields in fluid dynamics and electromagnetics.

Speaker notes